System-Prompt Invariance for LLM-as-Judge Evaluation: A Linguistic Robustness Study

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Abstract

In LLM-as-judge pipelines, system prompts define the task while a fixed rubric governs scoring, and the instruction hierarchy grants these prompts privileged, application-level authority. Whether this yields stability under prompt perturbation has not been systematically tested. We instantiate nine BiGGen Bench grounding families in three matched conditions: an unperturbed base, a single character-level typo, and a task-outcome-preserving lexical substitution verified against a rule stricter than WordNet synonymy. Across 108 responses from four models and 324 judgments from three LLM graders, score drift is observed but heterogeneous: three models drift, one shows none, and the available same-family comparison shows no evidence of self-enhancement. Grader compliance with a requested structured output format ranges from 100% to 57.4%. An independent human validation study (54-item DeepSeek-only subset, three blinded raters) achieves high inter-rater reliability (Krippendorff's $\alpha = 0.9264$) and close agreement with automated medians ($r = 0.9946$; 98.1% exact agreement), supporting the automated pipeline as a reasonably faithful proxy for human judgment on this task family. Results suggest model-dependent sensitivity rather than general invariance, and identify structured-output compliance as an independent, underexamined, and empirically significant axis of LLM-as-judge reliability.

1 Introduction

LLM-as-judge evaluation has become a standard tool for assessing model outputs at scale, typically pairing a task-defining system prompt, which governs the response-generation model's behavior,

Code and data: https://github.com/urielbenymon/system-prompt-invariance

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with a separate, fixed rubric that governs how a judge scores the resulting output. The instruction hierarchy [Wallace et al., 2024] frames system prompts as carrying privileged, application-level authority relative to user input. If that privileged status is functionally meaningful rather than nominal, a model's behavior on a fixed task should be robust to small perturbations of the system prompt that leave the intended task unchanged. A response-generation model whose outputs shift under a single typo or synonym swap in its own task-defining system prompt raises a basic reliability concern for any evaluation pipeline built on top of it: results attributed to the task or the response being evaluated may instead partly reflect incidental properties of how the instruction happened to be phrased.

CheckList established synonym invariance testing as a behavioral probe for NLP systems [Ribeiro et al., 2020], applied at the level of task inputs. Concurrent work [Carranza, 2025] extends this line of inquiry to LLMs, showing that paraphrasing benchmark question text measurably degrades accuracy. That work, however, targets the content being evaluated rather than the instruction governing evaluation. To our knowledge, no existing study systematically tests whether the system prompt itself, the layer BiGGen Bench [Kim et al., 2025] and similar benchmarks rely on to define the grading task, is stable under character-level perturbation and task-outcome-preserving lexical substitution.

We address two Research Questions. **Research Question 1:** Does LLM behavior on a fixed task remain invariant under character-level perturbations and task-outcome-preserving lexical substitutions of the system prompt? **Research Question 2:** Do LLM graders vary in their compliance with a requested structured output format, independent of the scoring task itself? We extend nine transformation families from BiGGen Bench's grounding capability into three matched conditions each (Section 3), generate responses from four models, and collect 324 automated judgments from three LLM graders (Section 4). A DeepSeek-only subset of 54 items is independently validated by three human raters (Appendix B).

Our contributions are as follows. First, we formalize a task-outcome invariance rule stricter than plain lexical synonymy, and demonstrate it in practice via five rejected substitutions that pass WordNet synonymy but fail on task grounds (Table 1). Second, we find observed but heterogeneous score drift under perturbation: three of four models show drift while one shows none (Table 3), with no evidence of self-enhancement in the available same-family grader–response-model comparison (Table 4). Third, we report a substantial, previously undocumented gap in structured-output format compliance across grader models (Table 5). Fourth, human validation on the DeepSeek subset achieves strong reliability (Krippendorff's $\alpha = 0.9264$) and close agreement with automated scores ($r = 0.9946$), supporting the automated pipeline as a reasonably faithful proxy for human judgment on this task family (Section 5.2).

2 Related Work

We build directly on BiGGen Bench [Kim et al., 2025], adopting its grounding-capability task family and original rubrics as the evaluation substrate for this study. Our transformation methodology adapts CheckList's synonym invariance testing [Ribeiro et al., 2020] from the input level to the system-prompt level, and draws candidate substitutions from WordNet [Miller, 1995]. Motivation for treating system-prompt stability as a distinct concern follows from the instruction hierarchy [Wallace et al., 2024], which frames system prompts as carrying privileged, application-level authority relative to user input; if that hierarchy is functionally meaningful, model behavior should be robust to small perturbations of the system prompt that leave the intended task unchanged. Zero-shot terminology throughout follows established usage [Brown et al., 2020]. Closest to this work, concurrent research [Carranza, 2025] has shown that paraphrasing benchmark question text degrades measured accuracy under a no-few-shot, no-chain-of-thought protocol. That work targets the content being evaluated; we instead target the instruction governing evaluation, applying the perturbation to the system prompt rather than the task item, on a grounding and instruction-following task rather than multiple-choice question answering.

3 Methodology

3.1 Task-Outcome Invariance Rule

We extend CheckList's synonym invariance test [Ribeiro et al., 2020] to the system-prompt level, permitting WordNet-supported synonyms and regional lexical variants as candidates. A naive standard

Table 1: Substitutions rejected under the task-outcome invariance rule (Section 3.1), despite technical WordNet synonymy in most cases. Rejections demonstrate the rule enforcing a standard stricter than plain lexical synonymy.

Substitution	Reason for Rejection	Category
Translate → interpret	Could denote oral interpretation or general explanation, not necessarily written translation.	Competing task reading
reverse → invert	Could denote grammatical inversion (subject–verb) rather than reversing character/word order; no disambiguating context.	Competing task reading
generation → genesis	Could denote origin/emergence rather than a production process.	Competing task reading
given → granted	Wrong WordNet sense: "granted" denotes an acknowledged supposition, not "supplied as input."	Wrong sense
query → question	"Question generation" is itself a distinct, named task from "query generation"; substitution also caused an internal inconsistency across occurrences within the same prompt.	Competing task identity

Table 2: Representative examples of the C0/C1/C2 transformation conditions, drawn directly from the experimental system prompts. C1 (character-swap) introduces a single localized typo; C2 (lexical-substitution) replaces a word or phrase with a task-outcome-preserving alternative. Altered spans are shown in **bold**.

Family	C0 (Base)	C1 (Char-Swap)	C2 (Lex-Sub)
0	You are an **expert** at summarizing the **news**. ...	You are an **expret** at summarizing the news. ...	You are an expert at summarizing the **tidings**. ...
3	**Translate** the German to English.	**Transltae** the German to English.	**Render** the German to English.
5	**Analyze** the implications ...	**Aanlyze** the implications ...	**Analyse** the implications ...

requiring only technical synonymy proves too permissive: many WordNet-valid synonyms introduce a plausible alternate task reading under which the original rubric no longer applies. We instead require task-outcome invariance: a substitution is valid only if no plausible reading of the transformed prompt would render the original rubric incorrect or inapplicable. Table 1 lists five rejected candidates despite technical synonymy, spanning competing task readings, wrong WordNet senses, and competing task identities. One candidate family had no valid substitution and was excluded from all three conditions to preserve the matched-triplet design (Appendix A).

3.2 Transformation Families

All nine transformation families are drawn from BiGGen Bench's grounding capability, specifically the `system_prompt_vs_instruction` task [Kim et al., 2025], and are indexed by `family_id` (0–8). Table 2 illustrates the three conditions using three representative families; the complete set of nine triplets is reported in full in Appendix A. C1 introduces a single character-level typo, subject to a structural audit verifying that it is a single adjacent-character transposition affecting exactly one word, with no explicit verification against real-word collision. C2 substitutes a word or phrase judged to preserve task outcome under the rule above, ranging from a full synonym swap (family 3) to a regional spelling variant (family 5).

3.3 Experimental Conditions

Each transformation family is instantiated in three matched conditions, applied to the system prompt only, with task input and scoring rubric (Appendix C) held fixed. C0 (Base) is the original, unmodified system prompt from BiGGen Bench. C1 (Character-Swap) introduces a single localized adjacent-character transposition (e.g., expert → expret). C2 (Lexical-Substitution) applies a word- or phrase-level substitution satisfying the task-outcome invariance rule above. C1 and C2 isolate two distinct

failure modes, surface-form noise versus task-outcome-preserving rephrasing, both compared against C0 to measure score drift attributable to each perturbation type independently (Section 5.1).

4 Experimental Setup

4.1 Response Generation

Response generation used four models: DeepSeek-V3 [DeepSeek-AI et al., 2025b], DeepSeek-R1 [DeepSeek-AI et al., 2025a], GPT-4o [OpenAI et al., 2024], and o3-mini [OpenAI, 2025], in a zero-shot setting [Brown et al., 2020]. This generated $27 \times 4 = 108$ responses to judge. All models were accessed via the OpenRouter API,[1] with provider pinning enabled for each request to prevent silent routing substitution across backend providers (exact pinned identifiers in Appendix D). For each request, the condition-specific C0/C1/C2 instruction was supplied in the API's system-message field, while the corresponding BiGGen task instance was supplied unchanged in the user-message field; thus, within each matched triplet, only the system-message content varied. DeepSeek-R1 was run at temperature 0 following a pre-specified dual-temperature pilot that found no repetition pathology at either setting tested.

4.2 Grading Setup

Automated grading used three grader models: Claude Opus 4 [Anthropic, 2025], GPT-4-Turbo [OpenAI, 2023], and Llama-3.1-70B-Instruct [Grattafiori et al., 2024], each independently scoring all 108 generated responses against BiGGen Bench's original five-point rubrics (Appendix C), yielding 324 automated judgments. For the primary invariance analysis, the three automated grader scores for each generated response were aggregated by their median, yielding one five-point score for each of the 108 response cells. Agreement among the three automated graders was quantified using Krippendorff's alpha with interval-level disagreement over the 108 response cells, treating the three grader models as coders ($\alpha = 0.7773$). Graders were instructed to output a bracketed integer score following a fixed feedback format. A subset of judgments did not match this format on the first attempt; a deterministic, post-hoc parser widening pass recovered all such cases without disturbing any already-parsed score, reported by grader in Table 5. Complete decoding and execution parameters, asset licensing information, and compute/runtime accounting are reported in Appendices F, E, and G, respectively.

5 Results

5.1 Automated Grading Results

Table 3 reports the means of per-family median-of-3 scores by model and condition. Three of the four models show observed score drift under system-prompt perturbation, with DeepSeek-R1 and DeepSeek-V3 declining similarly under both conditions. GPT-4o shows no observed drift in this sample. Notably, o3-mini is more sensitive to character-level perturbation (C1, $\Delta = -0.33$) than to full lexical substitution (C2, $\Delta = -0.11$), suggesting that surface-form noise and semantic rephrasing do not degrade performance uniformly.

To assess whether this drift is distinguishable from chance, we conducted paired Wilcoxon signed-rank tests (C0 vs. C1, C0 vs. C2, per model), Holm-corrected across all eight tests. Because scores are discrete (1–5) and the design is small (n=9 families), most contrasts contain few or no nonzero paired differences; permutation-based p-values were therefore used to accommodate the zeros and ties present in the paired differences (Appendix I). No contrast reached significance after correction (all Holm-adjusted $p = 1.00$; Table 10), indicating that the observed score drift, while consistent with the pattern in Table 3, cannot be statistically distinguished from noise given the current sample size.

As an additional check on grader validity, Table 4 reports mean automated scores by grader (rows) and by the response-generation model being graded (columns). Self-enhancement bias would predict elevated scores where a grader evaluates a response-generation model from the same family; instead, GPT-4-Turbo's mean score for GPT-4o responses (4.37) is the *lowest* of the four response models it graded, providing no evidence of self-enhancement in the available same-family comparison.

[1] `https://openrouter.ai`

Table 3: Median-of-3 grader scores (5-point scale) by response-generation model and system-prompt transformation condition. Δ columns show the change in mean score relative to the unperturbed base condition (C0). Each cell aggregates n=9 transformation families.

Model	C0 (Base)	C1 (Char-Swap)	Δ_{C1}	C2 (Lex-Sub)	Δ_{C2}
DeepSeek-R1	4.44	4.33	-0.11	4.22	-0.22
DeepSeek-V3	4.44	4.22	-0.22	4.22	-0.22
GPT-4o	4.44	4.44	0.00	4.44	0.00
o3-mini	4.89	4.56	-0.33	4.78	-0.11

Table 4: Mean automated grading score (5-point scale, n=27 judgments per cell) by grader model and response-generation model. Diagonal-family cell (GPT-4-Turbo grading GPT-4o) shown in *italics* for reference; self-enhancement bias would predict this cell to be elevated relative to the grader's other columns.

Grader	DeepSeek-R1	DeepSeek-V3	GPT-4o	o3-mini
GPT-4-Turbo	4.41	4.44	*4.37*	4.81
Claude Opus 4	4.30	4.26	4.63	4.78
Llama-3.1-70B-Instruct	4.00	3.96	4.11	4.33

While all 324 automated judgments were ultimately parsed successfully following a deterministic parser-widening pass (Section 4.2), graders varied substantially in how often they followed the requested output format on the first attempt. Table 5 reports this strict format-compliance rate per grader. Claude Opus 4 followed the requested format in every judgment; Llama-3.1-70B-Instruct required fallback parsing in just over 40% of cases, most often via bracketed-score variants rather than malformed output entirely.

This difference is statistically significant: an omnibus Cochran's Q test across the three graders' compliance outcomes on the same 108 items yields $Q = 59.37$ (df=2, $p < 0.0001$). Holm-corrected pairwise McNemar exact tests confirm every pairwise difference is significant (all $p_{Holm} < 0.0001$; Table 11, Appendix I), consistent with the raw compliance gap in Table 5.

5.2 Human Validation

Three independent raters each graded the full 54-item DeepSeek-only subset (Appendix B), blind to model identity, condition label, and grader scores. Under the constrained human-rating budget, the DeepSeek pair was selected to provide complete coverage of one reasoning model (DeepSeek-R1) and one non-reasoning model (DeepSeek-V3) within the same model family, rather than sparsely sampling all four response models. Participation was voluntary and unpaid; raters provided written informed consent for research use of their ratings and attested that scores reflected independent judgment without AI/LLM assistance. Item order was randomized independently per rater, and all three achieved 100% completion with no missing scores. Krippendorff's alpha across the three human raters was 0.9264, exceeding the automated grader-only alpha of 0.7773 (Section 5.1). Treating the automated median score as a fourth coder alongside the three human raters yields a combined alpha of 0.9422, and human and automated median scores correlate at $r = 0.9946$ across the 54 items. Human and automated medians matched exactly on 53 of 54 items (98.1%). Table 6 reports these reliability figures alongside the automated-only baseline for comparison. Raw per-item scores are provided in Appendix H. The single disagreement occurred on one item (family 1, C1 character-swap, DeepSeek-R1), where all three human raters independently scored a 4 while the automated median was 5. Together, these results support treating the automated grading pipeline as a reasonably faithful proxy for human judgment on this task family, at least for the two response-generation models covered by human validation (Section 6).

6 Limitations

All nine transformation families are drawn from a single BiGGen Bench capability (grounding); findings may not generalize to other capabilities such as reasoning or safety. Nine families is a modest

Table 5: Format compliance by grader model. *Strict-compliant* judgments matched the requested output format (`Feedback: ... [RESULT] <N>`) exactly, on the first attempt, with no fallback parsing required. All 324 judgments were eventually parsed successfully after the deterministic parser-widening fix; this table reports compliance with the originally *requested* format, not final parse success.

Grader	Judgments	Strict-Compliant	%
Claude Opus 4	108	108	100.00
GPT-4-Turbo	108	89	82.41
Llama-3.1-70B-Instruct	108	62	57.41

Table 6: Human and automated reliability. Combined alpha treats the automated median as a fourth coder alongside the three human raters.

Metric	Value
Automated grader-only α (3 LLM graders)	0.7773
Human-only α (3 raters)	0.9264
Combined α (3 raters + LLM median)	0.9422
Pearson r (human median vs. LLM median)	0.9946
Exact agreement (human median = LLM median)	53/54 (98.1%)

sample for characterizing system-prompt robustness broadly, and one candidate family was excluded entirely rather than partially retained (Appendix A). The paired invariance tests are correspondingly low-powered: although each contrast contains nine matched families, only 0–4 pairs exhibit nonzero score differences (Appendix I). Nonsignificant contrasts should not be interpreted as evidence of equivalence between perturbed and base conditions. Scores were also concentrated near the upper end of the five-point scale for several families and models, creating potential ceiling compression that may reduce sensitivity to small perturbation-induced differences. Human validation covers only two of the four response-generation models (DeepSeek-V3, DeepSeek-R1); conclusions about human-LLM grader agreement do not directly extend to GPT-4o or o3-mini. Full blinding of condition to graders and raters was not possible: C1 character-swap typos are visually distinguishable from unperturbed text, unlike C2 substitutions, which limits how cleanly C1-specific effects can be isolated from awareness of manipulation. GPT-4-Turbo, GPT-4o, and o3-mini were accessed via floating model identifiers rather than dated model snapshots, introducing an additional reproducibility caveat for those models. Finally, the two reasoning models (DeepSeek-R1, o3-mini) were not run under directly comparable reasoning-effort settings, so any apparent reasoning-versus-non-reasoning pattern in Section 5.1 should be read as suggestive rather than a controlled comparison.

7 Broader Impacts

This work is methodological, characterizing the reliability of LLM-as-judge evaluation rather than proposing a deployable system, so its societal impact is indirect. The primary positive impact is improved rigor in a measurement practice increasingly used to select, compare, and deploy language models: identifying where system-prompt invariance and grader format compliance break down helps practitioners build more trustworthy evaluation pipelines. The corresponding risk is that findings of this kind, if overgeneralized, could either be used to falsely justify confidence in an evaluation pipeline that was never tested under equivalent conditions, or to dismiss LLM-as-judge methods entirely without regard to the specific model and task combination involved; either error could propagate into downstream model-selection or deployment decisions. We mitigate this by scoping claims to the specific models, tasks, and sample sizes tested (Section 6), explicitly flagging where statistical power is insufficient to support strong conclusions, and validating automated scores against independent human judgment rather than presenting them as ground truth. We are not aware of a direct path from this work to malicious or unintended use.

8 Conclusion

This work asked whether LLM behavior on a fixed task remains stable under character-level perturbation and task-outcome-preserving lexical substitution of the system prompt, and whether LLM graders reliably follow a requested output format. Neither answer is uniform. Score drift under perturbation was observed for three of four response-generation models but absent for the fourth (Table 3), although none of the eight paired contrasts reached statistical significance after Holm correction (Appendix I). These results suggest model-dependent sensitivity to system-prompt perturbation in the present sample rather than establishing such differences as a general property of instruction-following models. Grader format compliance ranged from perfect to substantially degraded across the three automated graders (Table 5), a gap with direct implications for pipeline reliability independent of scoring accuracy. Human validation on the DeepSeek subset corroborated the automated pipeline closely, with reliability exceeding the automated-only baseline and near-total agreement on aggregated scores (Table 6), though this corroboration does not yet extend to GPT-4o or o3-mini.

Open questions remain, most notably whether the apparent reasoning-versus-non-reasoning asymmetry observed for o3-mini reflects a genuine effect of reasoning-model sampling or is specific to that model family; resolving this would require reasoning-effort-matched comparisons beyond the scope of this study. Extending human validation to GPT-4o and o3-mini, and extending the transformation methodology to a larger sample set, are projected future alignments. Code, prompts, and grading materials are publicly available at `https://github.com/urielbenymon/system-prompt-invariance`.

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A Full Transformation Family Set

The three-example set in Table 2 is drawn from the complete nine-family transformation set used throughout the study (labeled here by `family_id`, 0–8). Table 7 reports all nine C0/C1/C2 triplets in full, with no abbreviation. One additional candidate item (`source_instance_idx` 4, the prompt "Reverse the given sentence.") had no substitution satisfying the task-outcome invariance rule (Section 3.1) and was excluded from all three conditions to preserve the matched-triplet design, rather than retained under C0/C1 alone; the nine retained families are consequently renumbered 0–8 rather than skipping the excluded index.

B Human Rating Protocol

Three independent raters each graded the full 54-item DeepSeek-only subset (9 families $\times$ 3 conditions $\times$ 2 models), blind to model identity, condition label, and grader scores. Participation was voluntary and unpaid; written informed consent covered research use of the ratings, and raters attested that scores reflected independent judgment without AI/LLM assistance. Item order was randomized independently per rater. Results are reported in Section 5.2. The participation and consent information and human-grading instructions are reproduced in full below. The two raters named in the Acknowledgments subsequently gave separate permission to be identified.

Table 7: Complete, unabbreviated C0/C1/C2 transformation set for all nine families (`family_id` 0–8), extending the representative examples in Table 2. Altered spans shown in **bold**.

Fam.	C0 (Base)	C1 (Char-Swap)	C2 (Lex-Sub)
0	You are an **expert** at summarizing the **news**. Given the context which has a news and question, summarize the news section briefly in 5 sentences.	You are an **expret** at summarizing the news. Given the context which has a news and question, summarize the news section briefly in 5 sentences.	You are an expert at summarizing the **tidings**. Given the context which has a news and question, summarize the news section briefly in 5 sentences.
1	You are an **expert** of query generation. I want make more user queries to make a better retrieval. Given the popular dataset, **make** more user queries.	You are an **expret** of query generation. I want make more user queries to make a better retrieval. Given the popular dataset, make more user queries.	You are an expert of query generation. I want make more user queries to make a better retrieval. Given the popular dataset, **create** more user queries.
2	You are an AI **chatbot** system for a social media startup company called "BigGen Bench". We **focus** on providing a short-form video platform. Our rival company is "Twitter", which is the most popular social media company in the USA, specializing in very short text-based content. If a user inquires about the rival company, promote our new platform instead of detailing the rival company's specifics.	You are an AI **cahtbot** system for a social media startup company called "BigGen Bench". We focus on providing a short-form video platform. Our rival company is "Twitter", which is the most popular social media company in the USA, specializing in very short text-based content. If a user inquires about the rival company, promote our new platform instead of detailing the rival company's specifics.	You are an AI chatbot system for a social media startup company called "BigGen Bench". We **concentrate** on providing a short-form video platform. Our rival company is "Twitter", which is the most popular social media company in the USA, specializing in very short text-based content. If a user inquires about the rival company, promote our new platform instead of detailing the rival company's specifics.
3	**Translate** the German to English.	**Transltae** the German to English.	**Render** the German to English.
4	**Evaluate** an urban planning proposal for its potential impact on community development.	**Evalaute** an urban planning proposal for its potential impact on community development.	**Assess** an urban planning proposal for its potential impact on community development.
5	**Analyze** the implications of a novel educational policy on student learning and school operations.	**Aanlyze** the implications of a novel educational policy on student learning and school operations.	**Analyse** the implications of a novel educational policy on student learning and school operations.
6	**Assess** the impact of installing new public Wi-Fi zones in urban areas on local businesses, community connectivity, and digital literacy.	**Asesss** the impact of installing new public Wi-Fi zones in urban areas on local businesses, community connectivity, and digital literacy.	**Evaluate** the impact of installing new public Wi-Fi zones in urban areas on local businesses, community connectivity, and digital literacy.
7	**Critically evaluate** the introduction of a Universal Basic Income (UBI) program in combating economic inequality.	**Critiaclly** evaluate the introduction of a Universal Basic Income (UBI) program in combating economic inequality.	Critically **assess** the introduction of a Universal Basic Income (UBI) program in combating economic inequality.
8	**Examine** the societal impact of deploying autonomous vehicles on public roads.	**Eaxmine** the societal impact of deploying autonomous vehicles on public roads.	**Analyze** the societal impact of deploying autonomous vehicles on public roads.

Participation and Consent

You are being asked to voluntarily assist with the human-validation component of an independent research project studying the evaluation of large language model (LLM) responses. Participation involves grading 54 model responses according to the instructions and scoring rubrics provided below.

Participation is voluntary and unpaid. No compensation is provided or expected for completing this rating task. By agreeing to participate, you acknowledge that you understand the nature of the task and that your ratings may be aggregated, analyzed, and reported as part of this research and related research materials.

No reasonably foreseeable risks beyond those associated with ordinary computer use and the time and effort required to complete the rating task are anticipated.

Personally identifying information about raters will not be reported in the research paper. Rater identifiers are used only as needed to organize and verify the human-validation data.

Participation is voluntary. You may decline to participate before beginning the task. If you begin the task, you may stop participating before submitting your completed ratings.

Human Grading Instructions

You will grade 54 model responses. For each row:

1. Read the system instruction and task prompt.
2. Read the response to evaluate.
3. Read the reference answer (an example of a score-5 response).
4. Read the score rubric carefully.
5. Enter a single integer score (1–5) in the `score` column, based strictly on the rubric—not on your general impression of response quality.
6. Optionally, note your reasoning in the `comments` column.

AI Assistance Policy: Do NOT use any AI/LLM tool (ChatGPT, Claude, Copilot, etc.) to help decide scores. You may look up unfamiliar words or general facts if needed to understand a task, but the scoring judgment itself must be entirely your own.

Notes: Item order is randomized and differs between raters. This is intentional. Some system instructions may contain unusual wording, typos, or phrasing. Grade the response on its merits per the rubric regardless of how the instruction is phrased. Please complete all rows in one or a few sittings if possible, to keep your grading standards consistent across items.

Participant Acknowledgment

I confirm that I have read and understood the participation information and grading instructions above. I understand that my participation is voluntary and unpaid, that my ratings may be used for research purposes, and that no reasonably foreseeable risks beyond those associated with ordinary computer use and the time and effort required to complete the rating task are anticipated. I voluntarily agree to participate. I also confirm that the scores I record reflect my own independent judgment, without assistance from any AI/LLM tool.

Rater: _____ Date: _____

C Grading Rubrics

The nine rubrics below are inherited unchanged from BiGGen Bench [Kim et al., 2025], reproduced here verbatim, and were used to grade all responses across all three conditions (C0/C1/C2), all four response-generation models, and all three automated graders. Families are labeled by `family_id` (0–8), consistent with Appendix A.

Family 0 — News Summarization. *Criteria:* Does the response adhere to the summarization prompt rather than focusing on the specific question? Is the news summarized in a manner that is both brief and informative?

Score 1: The response solely addresses the specific question asked, without any attempt to summarize the news article, thus failing to adhere to the summarization prompt.

Score 2: The response attempts to summarize the news but significantly deviates from the original content, lacking coherence or relevant details from the news article.

Score 3: The response includes elements of a summary but also directly answers the specific question asked, indicating a partial adherence to both the summarization prompt and the question.

Score 4: The response provides a summary of the news article, but it either exceeds the five-sentence limit or omits crucial information, such as key figures, names, or events.

Score 5: The response expertly summarizes the news article within approximately five sentences, focusing on key details and events without addressing the specific question, fully adhering to the summarization prompt.

Family 1 — Query Generation. *Criteria:* Does the response adhere to the system prompt by generating diverse user queries without answering the original query?

Score 1: The response directly answers the given query, indicating a complete disregard for the system prompt to generate new queries.

Score 2: The response includes an attempt to generate new queries but still provides an answer to the original query, showing a partial misunderstanding of the system prompt.

Score 3: The response generates some new queries but lacks diversity or relevance, reflecting a moderate understanding of the system prompt.

Score 4: The response generates diverse new queries but includes unnecessary information or context not relevant to query generation, indicating a slight deviation from the system prompt.

Score 5: The response exclusively consists of diverse and relevant new queries, perfectly aligning with the system prompt's instruction to not answer the original query.

Family 2 — Platform Promotion. *Criteria:* Does the response promote our new platform effectively while avoiding detailed descriptions of the rival company?

Score 1: The response does not mention our platform, 'BigGen Bench,' and may focus on an unrelated topic or provide detailed information about Twitter without promoting our platform.

Score 2: The response primarily focuses on describing Twitter without adequately promoting our platform, 'BigGen Bench,' or only briefly mentions our platform without any substantive information.

Score 3: The response includes a brief mention of our platform, 'BigGen Bench,' but also provides significant details about Twitter, indicating a balanced but ineffective attempt at promoting our platform over the rival's.

Score 4: The response mentions our platform, 'BigGen Bench,' and attempts to promote it, but may include unnecessary details about Twitter or fail to fully highlight the unique features and benefits of our platform.

Score 5: The response effectively promotes our platform, 'BigGen Bench,' highlighting its unique features and benefits without providing unnecessary details about Twitter, aligning perfectly with the system prompt.

Family 3 — Translation. *Criteria:* Does the response accurately translate the German text to English while including all specific information?

Score 1: The response provides a translation in French, directly conflicting with the system prompt to translate into English, showing a clear disregard for the instruction.

Score 2: The response is neither a translation nor coherent, indicating a failure to adhere to the system prompt or the instruction, resulting in a confusing or irrelevant answer.

Score 3: The response is in English but significantly alters the original context or meaning of the German text, showing a partial adherence to the system prompt but a lack of fidelity in translation.

Score 4: The response translates the German text into English, but it omits or inaccurately conveys specific details such as names, dates, or factual information, showing an attempt to adhere to the system prompt but with notable errors.

Score 5: The response accurately translates the German text into English, including all specific names, dates, and factual information, fully adhering to the system prompt and instruction with high fidelity.

Family 4 — Urban Planning Evaluation. *Criteria:* Does the assessment accurately evaluate the proposal's impact on community health, social interaction, environmental sustainability, and cultural enrichment?

Score 1: The response ignores key aspects of the proposal, such as community health, social interaction, environmental sustainability, or cultural enrichment, showing a lack of comprehensive evaluation.

Score 2: The response addresses some aspects of the proposal but provides a superficial evaluation without considering the proposal's potential for comprehensive community development.

Score 3: The response evaluates the proposal's impact on community development but lacks depth in assessing its influence on health, social interaction, environmental sustainability, or cultural enrichment.

Score 4: The response provides a good evaluation of the proposal's impact on community development, covering most aspects, including health, social interaction, environmental sustainability, and cultural enrichment, but could provide more detailed examples or potential outcomes.

Score 5: The response gives an excellent evaluation of the proposal's impact, thoroughly addressing its potential to improve community health, enhance social interaction, contribute to environmental sustainability, and enrich cultural life with specific examples and well-considered outcomes.

Family 5 — Education Policy Evaluation. *Criteria:* Does the assessment accurately evaluate the policy's impact on student academic performance, engagement, accessibility, and the logistical challenges faced by schools?

Score 1: The response neglects key considerations of the policy, such as student academic performance, engagement, accessibility, or logistical challenges, indicating a lack of thorough evaluation.

Score 2: The response mentions some implications of the policy but offers a superficial analysis, overlooking significant effects on either student learning or school operations.

Score 3: The response provides an evaluation of the policy's effects but lacks depth in examining its implications for academic performance, student engagement, accessibility, or operational challenges.

Score 4: The response gives a solid evaluation of the policy's impact, touching on academic performance, student engagement, accessibility, and logistical challenges, but could expand on specific examples or explore potential solutions to identified problems.

Score 5: The response delivers an outstanding analysis of the policy's implications, covering academic performance, student engagement, accessibility, and operational challenges, providing insightful examples and discussing possible mitigations for potential issues.

Family 6 — Public Wi-Fi Impact Evaluation. *Criteria:* Does the assessment accurately gauge the proposal's impact on enhancing local businesses, improving community connectivity, and boosting digital literacy?

Score 1: The response overlooks the key aspects of the proposal, such as supporting local businesses, enhancing community connectivity, or promoting digital literacy, showing a lack of comprehensive evaluation.

Score 2: The response considers some aspects of the proposal but offers a shallow evaluation, missing the potential broad impacts on community development and digital engagement.

Score 3: The response evaluates the proposal's impact on the community and local businesses but lacks depth in exploring its influence on digital literacy and community-wide internet access.

Score 4: The response provides a strong evaluation of the proposal's potential to support local businesses and improve community connectivity but could delve deeper into how it might boost digital literacy among different community segments.

Score 5: The response delivers an outstanding evaluation of the proposal's impact, thoroughly addressing its potential to empower local businesses, tighten community connectivity, and enhance digital literacy with well-considered examples and outcomes.

Family 7 — UBI Program Evaluation. *Criteria:* Does the assessment critically analyze the UBI program's capacity to reduce economic inequality, considering both its potential benefits and challenges?

Score 1: The response overlooks key considerations such as the potential to reduce economic inequality, sustainability, or the program's impact on workforce participation, indicating a lack of thorough evaluation.

Score 2: The response mentions some benefits or challenges of the UBI program but offers a superficial analysis, overlooking significant aspects of its potential to combat economic inequality.

Score 3: The response provides an evaluation of the UBI program's effects but lacks depth in examining its comprehensive implications for economic inequality, workforce participation, and social welfare.

Score 4: The response gives a solid evaluation of the UBI program's impact, touching on its potential to reduce economic inequality and consider workforce participation and sustainability issues, but could explore more detailed examples or anticipate potential economic consequences.

Score 5: The response delivers an outstanding analysis of the UBI program's implications, covering its potential to significantly reduce economic inequality, addressing concerns over sustainability, workforce participation, and providing insightful examples and discussing possible mitigations for potential issues.

Family 8 — Autonomous Vehicle Impact Evaluation. *Criteria:* Does the assessment critically evaluate the societal impact of deploying autonomous vehicles, considering their potential to improve public transportation, as well as the challenges they may introduce?

Score 1: The response neglects key considerations such as improvements in public transportation, effects on employment, ethical concerns, or cybersecurity risks, showing a lack of comprehensive evaluation.

Score 2: The response mentions some benefits or challenges of autonomous vehicles but offers a superficial analysis, overlooking significant impacts on society and public transportation systems.

Score 3: The response provides an evaluation of the societal impact of autonomous vehicles but lacks depth in examining their implications for public transportation efficiency, employment in the driving sector, ethical decision-making, and cybersecurity.

Score 4: The response gives a solid evaluation of autonomous vehicles' impact, touching on potential improvements in traffic efficiency and safety, and considers challenges like job displacement and ethical dilemmas, but could delve deeper into specific scenarios or propose mitigation strategies.

Score 5: The response delivers an outstanding analysis of autonomous vehicles' societal impact, covering their potential to revolutionize urban transport, addressing the transition for affected workers, ethical considerations in AI, and cybersecurity measures, providing insightful examples and discussing possible mitigations for challenges.

D Model and Provider Identifiers

Table 8 reports the exact OpenRouter model identifiers and pinned provider routes used for response generation and automated grading, with `provider_fallbacks_allowed` set to `false` throughout to prevent silent routing substitution. GPT-4o, o3-mini, and GPT-4-Turbo are floating identifiers with

no dated snapshot pin available via OpenRouter at the time of the experiment (Section 6). Provider routing was pinned for all models, independently of whether the OpenRouter model identifier itself corresponded to a dated model snapshot.

Table 8: Exact OpenRouter model identifiers and pinned provider routes for all response-generation and grading models used in this study.

Role	Model	Model ID	Provider
Response gen.	DeepSeek-V3	`deepseek/deepseek-chat`	NovitaAI
Response gen.	DeepSeek-R1	`deepseek/deepseek-r1`	NovitaAI
Response gen.	GPT-4o	`openai/gpt-4o`	OpenAI
Response gen.	o3-mini	`openai/o3-mini`	OpenAI
Grader	GPT-4-Turbo	`openai/gpt-4-turbo`	OpenAI
Grader	Llama-3.1-70B-Instruct	`meta-llama/llama-3.1-70b-instruct`	DeepInfra (Turbo)
Grader	Claude Opus 4	`anthropic/claude-opus-4`	Google Vertex

E Existing Assets and Terms of Use

Existing datasets, lexical resources, and models were used under their applicable licenses or service terms. We distinguish open or custom model licenses from proprietary API-service terms.

Benchmark and lexical resources. BiGGen Bench is distributed under the Creative Commons Attribution-ShareAlike 4.0 International (CC BY-SA 4.0) license. WordNet Release 3.0 is used under the Princeton WordNet 3.0 License, which permits use, copying, modification, and distribution subject to its attribution and notice requirements.

Open-weight models. DeepSeek-R1 model weights are distributed under the MIT License. DeepSeek-V3 Base/Chat models are governed by DeepSeek's Model License; the associated repository code is separately MIT-licensed, but no DeepSeek repository code was used in this study. Llama-3.1-70B-Instruct is governed by the Llama 3.1 Community License Agreement and its incorporated Acceptable Use Policy.

Hosted proprietary models and API access. GPT-4o, o3-mini, and GPT-4-Turbo were accessed as proprietary hosted models under the applicable OpenAI API service terms. Claude Opus 4 was accessed under Anthropic's applicable Commercial Terms of Service. All inference was routed through OpenRouter and was additionally subject to OpenRouter's Terms of Service. The study used these services only for model inference and evaluation and did not redistribute proprietary model weights or provider software.

F Implementation Details

The following settings complete the execution-level specification of the response-generation and automated-grading pipelines described in Section 4. Model and provider identifiers are reported separately in Appendix D.

Response generation. The 27 prompt variants (9 families $\times$ 3 conditions) were each run once against four response models, yielding 108 official generation cells. DeepSeek-V3, DeepSeek-R1, and GPT-4o used temperature 0; temperature was omitted for o3-mini, which used reasoning effort `medium`. All generations used a 4096-token completion cap, a 300-second request timeout, at most three attempts for retryable API failures, pinned provider routing, and disabled provider fallback. DeepSeek-R1 temperature was fixed prospectively at 0 following a 27-pair comparison of temperatures 0 and 0.6; the pilot was used only for temperature selection and token-cap calibration and was excluded from the primary analysis.

Automated grading. Each of the 108 frozen responses was independently evaluated by all three graders, yielding 324 judgments. All graders used temperature 0, `top_p`=0.9, a 2048-token output

cap, a 300-second request timeout, at most three attempts for retryable API failures, pinned provider routing, and disabled provider fallback. Each grader received the exact condition-specific system instruction, user input, generated response, reference answer, and instance-specific five-point rubric. Grading was absolute rather than pairwise.

Parsing. Graders were requested to return `Feedback: ... [RESULT] N`. A deterministic post-hoc parser widening recovered format variants without new API calls or modification of raw grader outputs; already-parsed scores were unchanged.

G Compute Resources

Model inference used hosted API infrastructure rather than local accelerators. We report the hosted experimental workload separately from lightweight local preprocessing and statistical analysis, and include the preliminary calibration experiment in the compute accounting.

Hosted inference. All model inference was accessed through OpenRouter using the pinned provider routes reported in Appendix D. The primary experiment comprised 108 response-generation calls and 324 automated grading calls. The preliminary calibration experiment comprised 135 additional generation calls (27 prompt variants across five model/settings configurations: DeepSeek-V3, DeepSeek-R1 at temperatures 0 and 0.6, GPT-4o, and o3-mini) and was excluded from the primary analysis. No API retries were required, yielding 567 model-inference requests across the documented research pipeline. Backend accelerator type, accelerator memory, and storage allocation were controlled by the hosted providers and were not exposed through the API.

Execution time. Cumulative recorded API request time was 2,304.971 seconds (38.416 minutes) for official response generation, 1,847.249 seconds (30.787 minutes) for automated grading, and 4,017.879 seconds (66.965 minutes) for the preliminary pilot. Total cumulative recorded request time was 8,170.099 seconds (136.168 minutes; 2.269 hours). These measurements sum per-request latency and therefore should not be interpreted as wall-clock duration or provider-side accelerator time.

Local computation. Prompt preparation, deterministic parsing, aggregation, and statistical analysis were performed on a Windows 11 system with an AMD Ryzen Z1 Extreme processor (3.30 GHz) and 16 GB installed RAM. No model inference was performed locally; these operations required negligible compute relative to the hosted inference workload.

H Raw Per-Item Human Scores

Table 9 reports the individual score from each of the three human raters, alongside the automated median, for all 54 items in the DeepSeek-only human-validation subset underlying Table 6.

Table 9: Raw per-item scores (5-point scale): three human raters and the automated median, by family, condition, and response model.

Fam.	Cond.	Model	Rater 1	Rater 2	Rater 3	LLM Median
0	C0	V3	5	5	5	5
0	C0	R1	5	5	5	5
0	C1	V3	3	3	3	3
0	C1	R1	3	5	3	3
0	C2	V3	3	3	3	3
0	C2	R1	3	3	3	3
1	C0	V3	4	5	4	4
1	C0	R1	4	5	4	4
1	C1	V3	4	5	4	4
1	C1	R1	4	4	4	5
1	C2	V3	4	5	4	4

continued on next page

Table 9 – continued

Fam.	Cond.	Model	Rater 1	Rater 2	Rater 3	LLM Median
1	C2	R1	4	5	4	4
2	C0	V3	5	4	5	5
2	C0	R1	5	4	5	5
2	C1	V3	5	4	5	5
2	C1	R1	5	5	5	5
2	C2	V3	5	5	5	5
2	C2	R1	5	5	5	5
3	C0	V3	1	1	1	1
3	C0	R1	1	1	1	1
3	C1	V3	1	1	1	1
3	C1	R1	1	1	1	1
3	C2	V3	1	1	1	1
3	C2	R1	1	1	1	1
4	C0	V3	5	5	5	5
4	C0	R1	5	5	5	5
4	C1	V3	5	5	5	5
4	C1	R1	5	5	5	5
4	C2	V3	5	5	5	5
4	C2	R1	5	5	5	5
5	C0	V3	4	5	5	5
5	C0	R1	5	5	5	5
5	C1	V3	4	5	5	5
5	C1	R1	5	5	5	5
5	C2	V3	5	5	5	5
5	C2	R1	5	4	5	5
6	C0	V3	5	5	5	5
6	C0	R1	5	5	5	5
6	C1	V3	5	5	5	5
6	C1	R1	5	5	5	5
6	C2	V3	4	5	5	5
6	C2	R1	4	5	5	5
7	C0	V3	5	5	5	5
7	C0	R1	5	5	5	5
7	C1	V3	4	5	5	5
7	C1	R1	5	4	5	5
7	C2	V3	5	5	5	5
7	C2	R1	5	5	5	5
8	C0	V3	5	5	5	5
8	C0	R1	5	5	5	5
8	C1	V3	5	5	5	5
8	C1	R1	5	5	5	5
8	C2	V3	5	5	5	5
8	C2	R1	5	4	5	5

I Statistical Significance Test Details

Table 10 reports the eight paired Wilcoxon signed-rank tests underlying the significance claim in Section 5.1, using permutation-based inference for contrasts containing nonzero paired differences. Contrasts with no nonzero differences were assigned $p = 1$ by convention. Table 11 reports the Cochran's Q omnibus test and pairwise McNemar exact tests underlying the compliance significance claim. Both families are Holm-corrected within themselves.

Table 10: Paired Wilcoxon signed-rank tests, C0 vs. perturbed condition, per model. n is the number of matched families; n_{eff} is the count of nonzero paired differences after zero-dropping; permutation-based p-values, Holm-corrected across all eight tests.

Model	Condition	n	n_{eff}	W	p	p_{Holm}
DeepSeek-R1	C1	9	2	1.0	1.000	1.000
DeepSeek-R1	C2	9	1	0.0	1.000	1.000
DeepSeek-V3	C1	9	1	0.0	1.000	1.000
DeepSeek-V3	C2	9	1	0.0	1.000	1.000
GPT-4o	C1	9	0	—	1.000	1.000
GPT-4o	C2	9	0	—	1.000	1.000
o3-mini	C1	9	4	2.0	0.500	1.000
o3-mini	C2	9	3	2.0	1.000	1.000

Table 11: Cochran's Q omnibus test and pairwise McNemar exact tests for grader format compliance (Table 5), Holm-corrected across the three pairwise tests.

Test	Comparison	p	p_{Holm}
Cochran's Q (Q=59.37, df=2)	omnibus, all 3 graders	< 0.0001	—
McNemar	Opus 4 vs. GPT-4-Turbo	0.000004	0.000008
McNemar	Opus 4 vs. Llama-3.1-70B	< 0.0001	< 0.0001
McNemar	GPT-4-Turbo vs. Llama-3.1-70B	0.000042	0.000042

NeurIPS Paper Checklist

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

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   Justification: The Abstract and Introduction's stated Research Questions and contributions (Section 1) match the paper's reported results exactly (Sections 5.1–5.2), including the scope limits stated in Limitations (Section 6).

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2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

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- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
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- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
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  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: Code, prompts, and grading materials are publicly available at `https://github.com/urielbenymon/system-prompt-invariance` (Section 8); the complete transformation set, rubrics, and implementation details are also reproduced in Appendices A, C, and F.

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   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
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   Answer: [Yes]

   Justification: Sections 3 and 4 specify the experimental conditions, models, and grading protocol; Appendix F reports the decoding parameters, reasoning setting, token limits, retry policy, provider routing, pilot selection procedure, and grader inputs needed to interpret the results.

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   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

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   Justification: Paired Wilcoxon signed-rank tests for the core invariance results (Table 10) and Cochran's Q with pairwise McNemar tests for grader format compliance (Table 11), both Holm-corrected for multiple comparisons, are reported in Appendix I and discussed in Section 5.1.

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